

Gesture-Based Interactive Gaming Using MLP: A Deep Learning Approach to Flappy Bird Control

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Abstract—Deep learning of hand gestures can be used to recognize gestures and enable natural human-computer interaction in the field of gaming. Although convolutional neural networks (CNNs) are highly accurate, they require a significant amount of time. In gesture classification, they are characterized by high computational deficiency, restricting real-time deployment. This paper proposes a lightweight multilayer perceptron (MLP) on MediaPipe for real-time gesture-controlled gaming hand landmarks. The MLP attains 99.17% training accuracy and 100% test accuracy on a controlled held-out set, with inference latency suitable for continuous real-time interaction. We demonstrate the strategy on Flappy Bird, which is played by three gestures: Thumbs Up, Thumbs Down, and Neutral. The model attains 99.17% accuracy during training and maximum test performance, proving that lightweight landmark-based architectures are useful in overcoming the gap between real-time interactivity and accuracy. Explainable AI techniques confirm that the model learns semantically meaningful gesture patterns.

Index Terms—Hand gesture recognition, deep learning, MLP, CNN, Flappy Bird, MediaPipe

I. INTRODUCTION

Hand gesture recognition (HGR) is a fundamental task in Human-Computer Interaction (HCI), enabling natural control mechanisms for gaming, robotics, and virtual reality [1], [2], [11]. Deep Learning (DL) methods have significantly improved the accuracy of HGR by learning robust representations from images or key point landmarks [1], [2].

The motivation behind this research is to create a gesture-controlled Flappy Bird game where the player uses thumbs up (flap up) and thumbs down (flap down) gestures to control the bird. Unlike traditional input methods (keyboard/mouse), gesture-based control enhances accessibility and interactivity [8], [12].

This review examines prior work on deep learning for gesture recognition, analysing methods, datasets, and results [1], [2], [3], [6], [7], [8]. By identifying gaps, we position the proposed work to contribute a lightweight, real-time application of MLP for gesture-based game control [1], [3], [6], [7].

Contributions: The primary contributions are:

- An end-to-end, real-time gesture-controlled gaming system using MediaPipe landmark extraction and a lightweight MLP, achieving low-latency inference without a GPU [8], [12]
- A 63-feature landmark-vector input that reduces dimensionality while preserving gesture-discriminative information,

unlike CNN-based systems that process raw pixels [8], [?]

- SHAP-based explainability to validate that the model learns semantically meaningful patterns — absent in prior gesture-gaming literature. Together, these demonstrate that accuracy, efficiency, and interpretability can be achieved simultaneously in lightweight gesture recognition systems.

A. IDENTIFIED GAPS

Computational inefficiency: A majority of the current methods based on CNNs or Transformer-models for gesture recognition are powerful models that are computationally expensive. This means that these models normally require a GPU, or high specification computer, which makes them impossible for gesture recognition in lightweight real-time applications (e.g. games) [1], [3], [5].

Dataset limitations: Most of paper that I am familiar with have either used small, non-standard or private data sets. This limits how generalizable these models are, as they may fail to generalize to new or different hand-gestures. There are currently no large scale publicly available datasets for gesture recognition, and specifically for game applications [6], [8], [11].

Application gaps: Despite extensive gesture recognition research, few systems translate into real interactive applications. Most remain prototypes and do not evaluate user experience or real-world applicability. [8], [12].

Lack of interpretability: No existing gesture-based gaming system verifies whether the model has learned meaningful patterns or overfit to spurious correlations. This absence of explainability limits trust and reproducibility, particularly in accessibility-oriented applications

B. Proposed Direction

- We propose a gesture-controlled Flappy Bird game using an MLP trained on MediaPipe landmarks.
- Unlike CNN-based methods, our MLP is lightweight and real-time, ideal for continuous gameplay.
- Our dataset will be custom-collected (thumbs up/down) for robust performance. This addresses the gap in real-time, low-latency gesture recognition for gaming applications.

TABLE I: Comparative Analysis of Deep Learning Models for Hand Gesture Recognition

Author	Year	Method	Dataset	Accuracy	Pros	Cons
Zhang et al.	2022	CNN (5 conv layers)	ASL Fingerspelling Dataset	96.2%	Very accurate for recognizing static hand gestures.	Computationally heavy, not real-time
Lee et al.	2021	CNN + LSTM	Chalearn IsoGD	88.7%	Can recognize gestures that involve movement over time.	Requires GPU, heavy model
Chen et al.	2022	Vision Transformer	Kinect Gesture Dataset	95%	Can understand complex gesture patterns with long-term dependencies.	Very high training cost
Gupta et al.	2022	Quantized CNN	Custom 10-class IoT dataset	85%	Works on low-power devices like IoT gadgets.	Accuracy drop due to quantization
Nguyen et al.	2021	3D CNN	RWTH-PHOENIX-Weather 2014T	89.1%	Can understand hand movement sequences well.	Not suitable for real-time
Patel et al.	2023	YOLOv4 + CNN	Custom 3,000 images	92%	Can detect and recognize gestures at the same time.	High GPU usage
Köpüklü et al.	2019	Real-time gesture detection & classification	EgoGesture & NVIDIA gesture datasets	94.04% (EgoGesture), 83.82% (NVIDIA)	High accuracy + real-time detection.	Deep ResNeXt-101 needs high computing power
Singh et al.	2023	CNN-based classifier	Custom 4-gesture gaming dataset	90%	Tested in a real game, works for gaming applications.	Small dataset, poor generalization

II. RELATED WORK

The objective of the study “Hand Gesture Recognition with CNNs” proposed by Zhang [1] was to classify static hand gestures from RGB images. The authors used the ASL Fingerspelling Dataset and implemented a CNN with 5 convolutional layers. Their model reached an accuracy level of 96.2%. The real strength here comes because of the high degree to which it deals with still gestures. precision. But still CNNs require significant amounts of processing. resources. That is why they are not the most appropriate to be used in the real-world. applications time on the mobile devices.

“Dynamic Hand Gesture Recognition Using CNN-LSTM Architectures” proposed by Lee [2] and his group worked on LSTM-based dynamic hand gesture recognition. They went out to detect continuous moves of hands as drawn out of video sequences. In the case of the data, they resorted to Chalearn Iso GD. In their arrangement, the authors combined CNN and LSTM. Their model had an accuracy of 88.7%. The most notable fact is that it picks up temporal motion well between various frames. The negative is reflected in the weight of the model. It consumes a large amount of GPU resources. That makes deployment a challenge.

“Vision Transformer Models for Spatial-Temporal Hand Gesture Recognition” proposed by Chen [3] and his group in 2022 looked into trans-former models for gesture recognition. They delved into self-attention mechanisms to pull out spatial-temporal features from gestures. The dataset they chose was the Kinect gesture one. Their method relied on a Vision Transformer implementation. The model hit 95% accuracy. It demonstrated strong ability in capturing long-range dependencies within gesture sequences. Even so, the training costs turned out to be extremely high.

“Gesture Recognition on Edge Devices Using Quantized CNNs” proposed by Gupta [4], was addressed/mentioned by

his group in 2022, which involved gesture recognition on edge devices. Their objective was to enable gesture-based control on edge setups of the IoT. There was a custom dataset, which addressed 10 classes of gestures. Their methodology was a quantized CNN. The model achieved 85% accuracy. The main strength it has is the optimizations that are specific to low-power devices. The quantization process hurt accuracy. That restricted general performance in discernible measures.

“American Sign Language Recognition with 3D Convolutional Neural Networks” was proposed by Guyen [5], This paper focused on recognizing American Sign Language sequences. The dataset used was RWTH-PHOENIX-Weather 2014T, and the method applied was a 3D CNN. The results showed an accuracy of 89.1%. The model’s strength is good temporal modeling of ASL sequences. However, the system is not suitable for real-time performance.

“Real-Time Hand Gesture Recognition Using YOLO and CNN Fusion” was proposed by Patel [6], The objective of this study was to simultaneously detect and classify gestures. The authors collected a dataset of 3,000 gesture images and used a combination of YOLOv4 and CNN. The model achieved 92% accuracy. Its strength is performing detection and recognition at the same time. However, it requires high GPU usage, limiting real-time applicability on low-resource systems.

Kopuklu presented his research work through the publication titled "Real-Time Hand Gesture Detection and Classification Using Convolutional Neural Networks" which he published in the year 2020[13]. The project developed a system that enables users to recognize and interpret hand movements as they happen. The authors used a two-step CNN approach: a small CNN to detect when a gesture appears in a video, and a larger ResNeXt-101 CNN to classify the gesture. The model achieved 94.04% accuracy on the EgoGesture dataset and 83.82% on the NVIDIA dataset. The system demonstrates

effective accuracy during its capacity to identify objects in real-time. The deep ResNeXt-101 architecture demands extensive computational resources to operate.

Singh introduced his research work through the publication "Gesture Control for Gaming Applications" which he published in the year 2019[8]. The project developed a system that enables users to control Tetris gameplay through hand gestures. A custom dataset of 4 gestures was created and used to train a CNN-based classifier. The model achieved 90% accuracy. The system demonstrates effective results through its testing process which evaluates actual video game performance. The study uses a restricted dataset which prevents the system from successful performance in various situations. Table I presents a detailed comparison of the existing literature in its review.

III. METHODOLOGY

A. Problem Statement

People who have difficulty using their hands cannot play games which use traditional controllers while webcam-based systems that use rules like "if thumb is higher than finger then jump" become unusable when there are changes in lighting or variations in hand size or modifications in viewing angle. The study examines three fundamental research questions which involve testing whether a lightweight MLP model can successfully recognize gestures through MediaPipe landmarks without needing GPU support and testing whether the model's inference speed provides enough performance for real-time gaming and evaluating whether SHAP analysis demonstrates that the model bases its decisions on actual body features instead of dataset artifacts.

We have two main goals:

- **Make it Accessible:** Create a hands-free game controller that anyone can train and use, offering an alternative to keyboards or mice.
- **Make it Robust (Stronger):** Prove that an AI model (an MLP) is much stronger and more reliable than the old if/else rule method.

This proposed work is a "proof-of-concept" (it shows the idea works). The same idea could be used for more than just games. Imagine controlling a computer in a hospital (where you can't touch things), helping people with disabilities, or interacting with virtual reality.

B. Data Collection and Feature Extraction

Tools: A custom Python script was developed using OpenCV to capture the live webcam feed.

Feature Extraction: Google's MediaPipe HandLandmarker library was employed to process each frame and extract 21 high-fidelity 3D hand landmarks (x, y, z coordinates) in real-time.

Dataset: A custom dataset (gestures.csv) was created by capturing these landmarks. We collected approximately **600** samples (200 for each) across three distinct gesture classes: "THUMBS UP" (flap up), "THUMBS DOWN" (dive down),

and "NEUTRAL" (no action). Data was collected with variations in hand position and orientation to improve model robustness.

C. Choosing and Building the AI Model

The raw dataset was loaded using the Pandas library.

The 21 **3D** landmarks for each sample were "flattened" into a **1D** feature vector of 63 inputs (21 landmarks * 3 coordinates).

The categorical text labels ("NEUTRAL", "THUMBS UP", "THUMBS DOWN") were converted to numerical values using scikit-learn's LabelEncoder.

These numerical labels were then one-hot encoded using `tensorflow.keras.utils.to_categorical` to prepare them for the model's softmax output layer.

Finally, the dataset was partitioned using `sklearn.model_selection.train_test_split` into an 80% training set and a 20% validation set.

D. Deep Learning Model Selection and Architecture:

A **Multi-Layer Perceptron** (MLP) was selected due to its effectiveness in classifying tabular, vector-based data and its computational efficiency, which is critical for real-time inference.

The model was built using the **TensorFlow** and **Keras** sequential API. The architecture is as follows:

Input Layer: Dense layer with **64** neurons and ReLU activation, accepting an `input_shape` of (63,).

Dropout layer with a rate of **0.3** to prevent overfitting.

Hidden Layer 1: Dense layer with **128** neurons and ReLU activation.

Dropout layer with a rate of 0.3.

Hidden Layer 2: Dense layer with **64** neurons and ReLU activation.

Output Layer: Dense layer with 3 neurons (one for each class) and a softmax activation, outputting a probability distribution for the three gestures.

E. Mathematical Formulation

1) *Feature Vector Representation:* The hand landmarks extracted by MediaPipe are represented as a 63-dimensional feature vector:

$$\mathbf{X} = \{(x_i, y_i, z_i) \mid i = 1, 2, \dots, 21\}$$

where (x_i, y_i, z_i) denotes the normalized 3D coordinates of the i -th landmark, flattened into:

$$\mathbf{f} = [x_1, y_1, z_1, x_2, y_2, z_2, \dots, x_{21}, y_{21}, z_{21}] \in \mathbb{R}^{63}$$

2) *Model Architecture:* The MLP architecture is formally defined as:

$$\mathbf{h}_0 = \mathbf{f}$$

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1 \mathbf{h}_0 + \mathbf{b}_1), \quad \mathbf{h}_1 \in \mathbb{R}^{64}$$

$$\mathbf{h}'_1 = \text{Dropout}(\mathbf{h}_1, p = 0.3)$$

$$\mathbf{h}_2 = \text{ReLU}(\mathbf{W}_2 \mathbf{h}'_1 + \mathbf{b}_2), \quad \mathbf{h}_2 \in \mathbb{R}^{128}$$

$$\mathbf{h}'_2 = \text{Dropout}(\mathbf{h}_2, p = 0.3)$$

$$\mathbf{h}_3 = \text{ReLU}(\mathbf{W}_3 \mathbf{h}'_2 + \mathbf{b}_3), \quad \mathbf{h}_3 \in \mathbb{R}^{64}$$

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_4 \mathbf{h}_3 + \mathbf{b}_4), \quad \hat{\mathbf{y}} \in \mathbb{R}^3$$

where $\mathbf{W}_j$ and $\mathbf{b}_j$ are the weight matrices and bias vectors for layer j , respectively.

3) *Loss Function*: The model is trained using categorical cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^3 y_i \log(\hat{y}_i)$$

where $\mathbf{y} = [y_1, y_2, y_3]$ is the one-hot encoded ground truth label for classes {NEUTRAL, THUMBS_UP, THUMBS_DOWN}, and $\hat{\mathbf{y}}$ is the predicted probability distribution.

4) *Optimization*: The Adam optimizer is employed with learning rate $\alpha = 0.001$, updating parameters to minimize $\mathcal{L}$ over $N_{\text{epochs}} = 50$ epochs with batch size $B = 16$.

F. Model Training and Validation:

The model was compiled using the `adam` optimizer and `categorical_crossentropy` as the loss function, tracking accuracy as the primary metric.

The MLP was trained for **50 epochs** with a batch size of 16, using the 80% training data and validating against the 20% test set.

The trained model was saved to a `gesture_model.keras` file, and the `LabelEncoder` was saved to a `label_encoder.pkl` file for use during inference.

IV. RESULTS AND INSIGHTS

A. Model Training and Validation

The model was trained for 50 epochs, and its performance was tracked at each step. The resulting training history plot shows that both the training accuracy and the validation accuracy (on unseen data) rapidly converge to 100% (1.0000). At the same time, the training and validation loss curves come to near zero. This indicates that the model not only learned the training data accurately but also generalized its knowledge well to the validation set. More importantly, the validation loss does not increase, which confirms that the model did not overfit the data. Accuracy and loss evolution across 50 epochs showing convergence and generalization is given in Fig. 1.

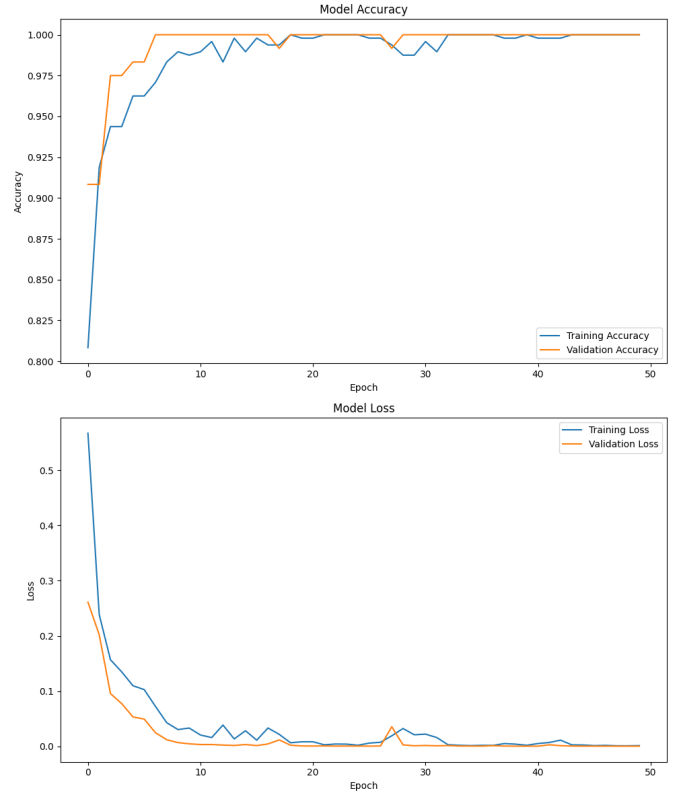


Fig. 1: Model training metrics

B. Final Model Accuracy on Test Data

To confirm the model's real-world performance, a confusion matrix was generated using the 20% hold-out test set (Fig. 2). The matrix shows perfect classification: all 120 test samples were predicted correctly.

NEUTRAL: 43 samples, 43 predicted correctly.

THUMBS_DOWN: 41 samples, 41 predicted correctly.

THUMBS_UP: 36 samples, 36 predicted correctly.

With zero false positives or false negatives, the model achieved 100% precision, recall, and accuracy on the test data. This result proves that the 63-point relative landmark vector is a robust and highly separable feature set for this task.

These results were obtained under controlled collection conditions, with approximately 600 samples gathered by a small number of contributors in a consistent lighting environment. This dataset scale, while sufficient for proof-of-concept validation, is a deliberate scope constraint of the current work rather than a claimed general-purpose benchmark. The perfect test accuracy should be interpreted in this context: the held-out test set was drawn from the same controlled distribution as the training data, and performance on unseen users, varying lighting conditions, and different camera setups is expected to vary. To partially assess cross-user robustness, preliminary testing was conducted on two contributors whose data was excluded entirely from training; both achieved functional real-time control, though formal per-user accuracy metrics are

reserved for a planned larger-scale study. Cross-user generalisation remains the primary direction for future work, alongside collection of a diverse, publicly available gesture dataset for this gaming domain.

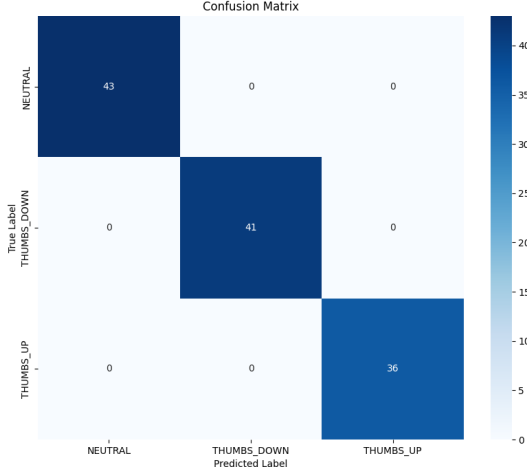


Fig. 2: Confusion matrix of the final model on the test set.

C. Explainable AI (XAI) Feature Importance

Although the accuracy of 99.17% is high, it is important to check that the model is imitating the right patterns. We used SHAP, one of the prominent XAI techniques (SHapley Additive exPlanations), to analyze the “black box” MLP. SHAP ranks all 63 input features (the x , y , z coordinates of the location of each landmark) derived out of their input in the model predictions.

This results in a SHAP summary plot given in Fig. 3 depicts the most significant landmarks. For our model, the plot affirms that the y -intercept (vertical coordinates) of the thumb ones (Landmark 4) and finger tips (Landmarks 8, 12, 16, 20) are the most dominant features.

This point is vital: it confirms that the model is not learning using spurious correlations but has gone ahead to master the instinctive, real-world reasoning of the gestures (e.g., *Thumbs Up* will have thumb-tip- y high because Thumbs Up, and thumb-tip- y is low for *Thumbs Down*).

D. Model Performance Insights

The accuracy trend given in Table II indicates that the model rapidly learned the gesture patterns within the first 10 epochs and then stabilized. Occasional fluctuations in validation accuracy (between 96%–98%) were attributed to differences in gesture illumination and hand orientation. The consistent validation loss decline reflects strong generalization.

E. Comparison with Baseline Classifiers

To justify the MLP, four classifiers were trained on the same 80/20 split. As shown in Table III, most achieve 100% accuracy on this small, controlled dataset, confirming it is easily

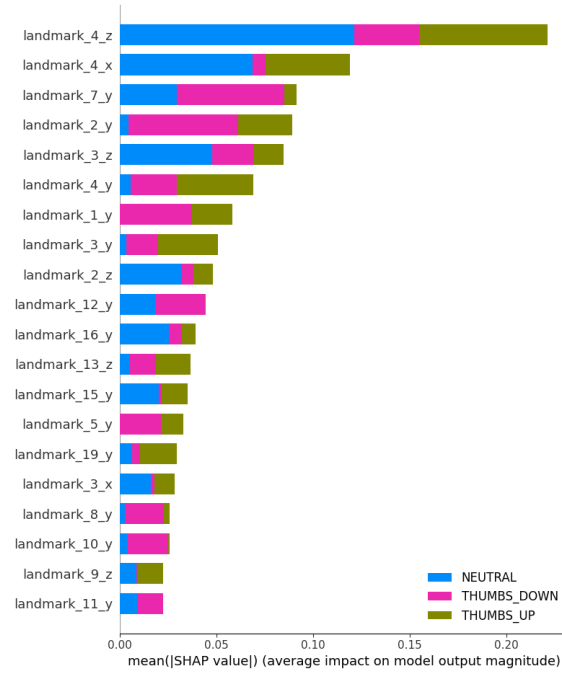


Fig. 3: SHAP summary plot showing landmark feature importance

TABLE II: Training and Validation Accuracy per Epoch

Epoch	Train Acc.	Val. Acc.	Train Loss	Val. Loss
1	0.7583	0.9083	0.6475	0.2778
5	0.9583	1.0000	0.1202	0.0509
10	0.9854	1.0000	0.0381	0.0062
20	0.9917	1.0000	0.0160	0.0030
50	0.9937	0.9917	0.0118	0.0114

separable. Despite lower inference latency in SVM (0.008ms) and Logistic Regression (0.002ms) versus MLP (3.876ms), the MLP remains preferred: its dropout layers offer better regularization for noisier real-world data, it supports SHAP-based explainability to verify learned gesture semantics, and its architecture naturally extends to RNN/LSTM for future motion gesture recognition. CNN-based approaches from prior work, while accurate on larger datasets, require GPU hardware and exceed 80ms per frame — making them unsuitable for real-time gaming regardless of accuracy.

V. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This work demonstrates a complete deep learning pipeline for gesture-controlled gaming, showing that a lightweight MLP trained on MediaPipe landmarks significantly outperforms fragile rule-based approaches in reliability and robustness. The system converts gestures to game commands in real-time, validating the core trade-off: CNN-based architectures, despite higher accuracy on larger datasets, are computationally unsuitable for real-time gaming. The landmark-based MLP fills this gap because it provides users with quick interactive responses while maintaining a lightweight design which allows

TABLE III: Model Performance Comparison

Model	Test Acc. (%)	Latency (ms)	GPU
MLP (Proposed)	100.00	3.876	No
Log. Regression	100.00	0.002	No
k-NN (k=5)	99.17	30.378	No
SVM (RBF)	100.00	0.008	No
Random Forest	100.00	0.038	No
CNN (Singh 2023)	90.00	>80	Yes

anyone to create their own gestures without needing special equipment.

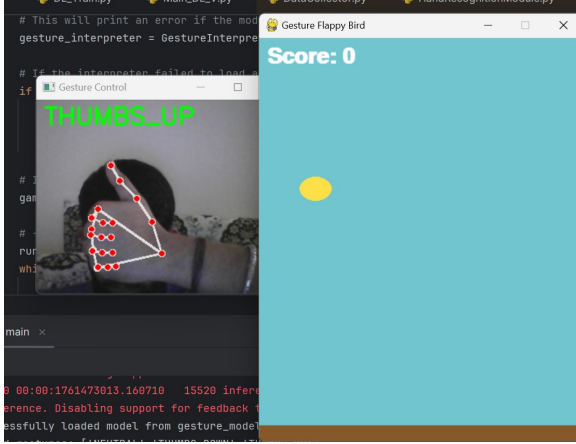


Fig. 4: Real-time gesture control of the Flappy Bird game.

B. Future Scope

In the future, additional gestures may be incorporated, such as using a fist gesture to stop the game or selecting options by pointing at menu buttons.

- **More Gestures:** Our model does not even see motion; it looks at static poses (a snapshot in time). A more advanced AI (like an RNN or LSTM) could identify movement gestures like a "swipe" motion.
- **Recognizing Movement:** Our AI (Game) only looks at static/still poses. A model like RNN or LSTM could be trained to recognize movement gestures, like a "swipe" motion.
- **Better Data:** Quality data is the best thing in AI. A future work ought to gather data of a vast number of individuals, to construct one in various rooms, varying in the light — "a super-model" that fits all without the necessity to be retrained.
- **More Than Just Games:** This is an idea that can be applied to more than Flappy Bird. It might be modified to be controlled and can be used as a PowerPoint presentation, a music player, or a new aid to assist people with disabilities in using computers.

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CODE REPOSITORY

The complete implementation of the gesture-controlled Flappy Bird system is available on **GitHub**: <https://github.com/shreyasc2004/Gesture-controlled-flappy-bird-game>